

```
import pandas as pd
import numpy as np

movies = pd.read_csv(r"C:\Users\HP\Downloads\archive\movie.csv")
```

```
movies
```

	movieId	title \
0	1	Toy Story (1995)
1	2	Jumanji (1995)
2	3	Grumpier Old Men (1995)
3	4	Waiting to Exhale (1995)
4	5	Father of the Bride Part II (1995)
...	...	...
27273	131254	Kein Bund für's Leben (2007)
27274	131256	Feuer, Eis & Dosenbier (2002)
27275	131258	The Pirates (2014)
27276	131260	Rentun Ruusu (2001)
27277	131262	Innocence (2014)

	genres
0	Adventure Animation Children Comedy Fantasy
1	Adventure Children Fantasy
2	Comedy Romance
3	Comedy Drama Romance
4	Comedy
...	...
27273	Comedy
27274	Comedy
27275	Adventure
27276	(no genres listed)
27277	Adventure Fantasy Horror

```
[27278 rows x 3 columns]
```

```
movies.shape
```

```
(27278, 3)
```

```
movies.columns
```

```
Index(['movieId', 'title', 'genres'], dtype='object')
```

```
movies.head()
```

	movieId	title \
0	1	Toy Story (1995)
1	2	Jumanji (1995)
2	3	Grumpier Old Men (1995)

```

3         4           Waiting to Exhale (1995)
4         5  Father of the Bride Part II (1995)

```

```

                                genres
0  Adventure|Animation|Children|Comedy|Fantasy
1                Adventure|Children|Fantasy
2                        Comedy|Romance
3                Comedy|Drama|Romance
4                        Comedy

```

```
tags =pd.read_csv(r"C:\Users\HP\Downloads\archive\tag.csv")
```

```
tags
```

```

      userId  movieId      tag      timestamp
0         18    4141  Mark Waters  2009-04-24 18:19:40
1         65     208   dark hero  2013-05-10 01:41:18
2         65     353   dark hero  2013-05-10 01:41:19
3         65     521  noir thriller  2013-05-10 01:39:43
4         65     592   dark hero  2013-05-10 01:41:18
...      ...      ...      ...      ...
465559  138446   55999   dragged  2013-01-23 23:29:32
465560  138446   55999  Jason Bateman  2013-01-23 23:29:38
465561  138446   55999   quirky  2013-01-23 23:29:38
465562  138446   55999    sad  2013-01-23 23:29:32
465563  138472     923  rise to power  2007-11-02 21:12:47

```

```
[465564 rows x 4 columns]
```

```
tags.shape
```

```
(465564, 4)
```

```
tags.columns
```

```
Index(['userId', 'movieId', 'tag', 'timestamp'], dtype='object')
```

```
tags.head()
```

```

      userId  movieId      tag      timestamp
0         18    4141  Mark Waters  2009-04-24 18:19:40
1         65     208   dark hero  2013-05-10 01:41:18
2         65     353   dark hero  2013-05-10 01:41:19
3         65     521  noir thriller  2013-05-10 01:39:43
4         65     592   dark hero  2013-05-10 01:41:18

```

```
ratings =pd.read_csv(r"C:\Users\HP\Downloads\archive\rating.csv")
```

```
ratings
```

```

      userId  movieId  rating      timestamp
0          1         2     3.5  2005-04-02 23:53:47

```

```

1          1          29          3.5  2005-04-02 23:31:16
2          1          32          3.5  2005-04-02 23:33:39
3          1          47          3.5  2005-04-02 23:32:07
4          1          50          3.5  2005-04-02 23:29:40
...      ...      ...      ...      ...
20000258  138493    68954    4.5  2009-11-13 15:42:00
20000259  138493    69526    4.5  2009-12-03 18:31:48
20000260  138493    69644    3.0  2009-12-07 18:10:57
20000261  138493    70286    5.0  2009-11-13 15:42:24
20000262  138493    71619    2.5  2009-10-17 20:25:36

```

```
[20000263 rows x 4 columns]
```

```
ratings.shape
```

```
(20000263, 4)
```

```
ratings.columns
```

```
Index(['userId', 'movieId', 'rating', 'timestamp'], dtype='object')
```

```
ratings.head()
```

```

   userId  movieId  rating  timestamp
0        1         2     3.5  2005-04-02 23:53:47
1        1        29     3.5  2005-04-02 23:31:16
2        1        32     3.5  2005-04-02 23:33:39
3        1        47     3.5  2005-04-02 23:32:07
4        1        50     3.5  2005-04-02 23:29:40

```

```
Data Structures
```

```
row_0 = tags.iloc[0]
```

```
type(row_0)
```

```
pandas.core.series.Series
```

```
print(row_0)
```

```

userId          18
movieId         4141
tag             Mark Waters
timestamp  2009-04-24 18:19:40
Name: 0, dtype: object

```

```
row_0.index
```

```
Index(['userId', 'movieId', 'tag', 'timestamp'], dtype='object')
```

```
row_0['userId']
```

```
np.int64(18)
```

```
'rating' in row_0
```

```

False
row_0.name
0
row_0 = row_0.rename('firstRow')
row_0.name
'firstRow'
DataFrames
tags.head()
   userId  movieId      tag      timestamp
0      18     4141  Mark Waters  2009-04-24 18:19:40
1      65      208   dark hero  2013-05-10 01:41:18
2      65      353   dark hero  2013-05-10 01:41:19
3      65      521  noir thriller  2013-05-10 01:39:43
4      65      592   dark hero  2013-05-10 01:41:18
tags.index
RangeIndex(start=0, stop=465564, step=1)
tags.columns
Index(['userId', 'movieId', 'tag', 'timestamp'], dtype='object')
tags.iloc[ [0,11,500] ]
   userId  movieId      tag      timestamp
0      18     4141  Mark Waters  2009-04-24 18:19:40
11     65     1783  noir thriller  2013-05-10 01:39:43
500   342    55908  entirely dialogue  2012-01-31 18:41:16
Descriptive Statistics
ratings['rating'].describe()
count    2.000026e+07
mean     3.525529e+00
std      1.051989e+00
min      5.000000e-01
25%      3.000000e+00
50%      3.500000e+00
75%      4.000000e+00
max       5.000000e+00
Name: rating, dtype: float64
ratings.describe()
   count  userId  movieId  rating
count  2.000026e+07  2.000026e+07  2.000026e+07

```

```

mean    6.904587e+04  9.041567e+03  3.525529e+00
std     4.003863e+04  1.978948e+04  1.051989e+00
min     1.000000e+00  1.000000e+00  5.000000e-01
25%     3.439500e+04  9.020000e+02  3.000000e+00
50%     6.914100e+04  2.167000e+03  3.500000e+00
75%     1.036370e+05  4.770000e+03  4.000000e+00
max     1.384930e+05  1.312620e+05  5.000000e+00

```

```
ratings['rating'].mean()
```

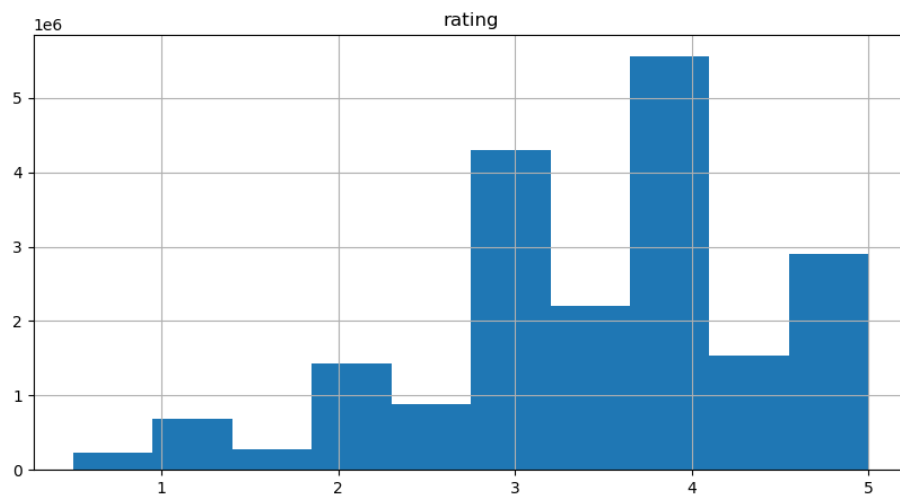
```
np.float64(3.5255285642993797)
```

Data Visualization

```
%matplotlib inline
```

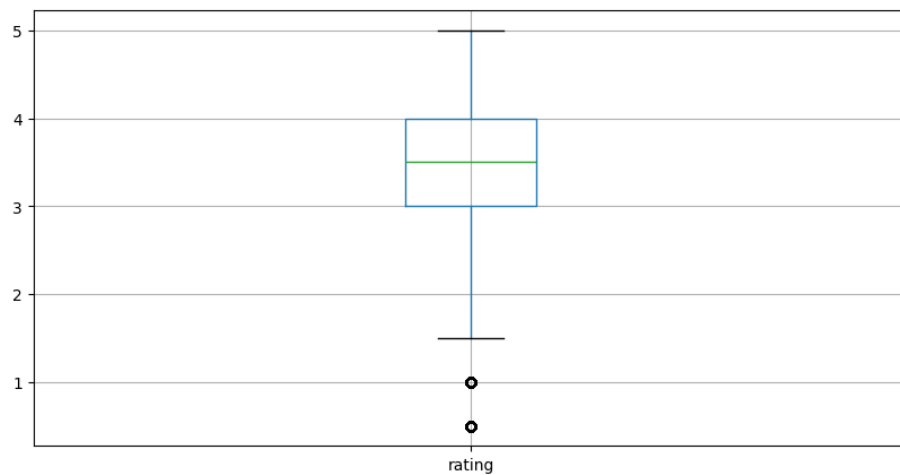
```
ratings.hist(column='rating', figsize=(10,5))
```

```
array([[<Axes: title={'center': 'rating'}>]], dtype=object)
```



```
ratings.boxplot(column='rating', figsize=(10,5))
```

```
<Axes: >
```



Slicing Out Columns

```
tags['tag'].head()
```

```
0    Mark Waters
1    dark hero
2    dark hero
3    noir thriller
4    dark hero
```

Name: tag, dtype: object

```
movies[['title', 'genres']].head()
```

```

                                title \
0                Toy Story (1995)
1                Jumanji (1995)
2        Grumpier Old Men (1995)
3        Waiting to Exhale (1995)
4  Father of the Bride Part II (1995)

```

```

                                genres
0  Adventure|Animation|Children|Comedy|Fantasy
1                Adventure|Children|Fantasy
2                        Comedy|Romance
3                Comedy|Drama|Romance
4                        Comedy

```

```
ratings[-10:]
```

```

      userId  movieId  rating      timestamp
20000253  138493    60816    4.5  2009-12-03 18:32:43
20000254  138493    61160    4.0  2009-11-16 16:55:37
20000255  138493    65682    4.5  2009-10-17 21:52:53

```

```

20000256 138493 66762 4.5 2009-10-17 18:50:08
20000257 138493 68319 4.5 2009-12-07 18:15:20
20000258 138493 68954 4.5 2009-11-13 15:42:00
20000259 138493 69526 4.5 2009-12-03 18:31:48
20000260 138493 69644 3.0 2009-12-07 18:10:57
20000261 138493 70286 5.0 2009-11-13 15:42:24
20000262 138493 71619 2.5 2009-10-17 20:25:36

```

```

tag_counts = tags['tag'].value_counts()
tag_counts[-10:]

```

```

tag
missing child          1
Ron Moore              1
Citizen Kane           1
mullet                1
biker gang             1
Paul Adelstein         1
the wig                1
killer fish            1
genetically modified monsters 1
topless scene          1
Name: count, dtype: int64

```

```

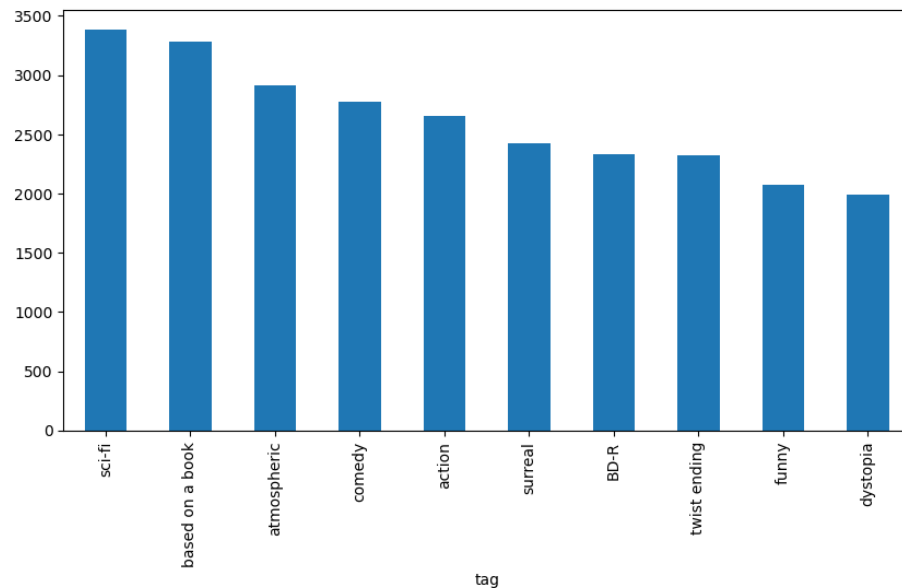
tag_counts[:10].plot(kind='bar', figsize=(10,5))

```

```

<Axes: xlabel='tag'>

```



Filters for Selecting Rows

```
is_highly_rated = ratings['rating'] >= 5.0
ratings[is_highly_rated][30:50]
```

	userId	movieId	rating	timestamp
239	3	50	5.0	1999-12-11 13:13:38
242	3	175	5.0	1999-12-11 13:32:13
244	3	223	5.0	1999-12-11 13:20:44
245	3	260	5.0	1999-12-11 13:09:02
246	3	316	5.0	1999-12-14 12:51:10
247	3	318	5.0	1999-12-11 13:09:26
248	3	329	5.0	1999-12-14 12:53:41
252	3	457	5.0	1999-12-11 13:16:55
253	3	480	5.0	1999-12-14 12:50:20
254	3	490	5.0	1999-12-11 13:30:41
256	3	541	5.0	1999-12-11 13:14:07
258	3	593	5.0	1999-12-11 13:16:55
263	3	858	5.0	1999-12-11 13:01:07
264	3	904	5.0	1999-12-11 13:04:10
267	3	924	5.0	1999-12-11 13:10:12
268	3	953	5.0	1999-12-11 13:11:52
271	3	1060	5.0	1999-12-11 13:18:30
272	3	1073	5.0	1999-12-11 13:20:44
275	3	1084	5.0	1999-12-11 13:15:03
276	3	1089	5.0	1999-12-11 13:05:58

```
is_action= movies['genres'].str.contains('Action')
movies[is_action][5:15]
```

	movieId	title \
22	23	Assassins (1995)
41	42	Dead Presidents (1995)
43	44	Mortal Kombat (1995)
50	51	Guardian Angel (1994)
65	66	Lawnmower Man 2: Beyond Cyberspace (1996)
69	70	From Dusk Till Dawn (1996)
70	71	Fair Game (1995)
75	76	Screamers (1995)
77	78	Crossing Guard, The (1995)
85	86	White Squall (1996)

	genres
22	Action Crime Thriller
41	Action Crime Drama
43	Action Adventure Fantasy
50	Action Drama Thriller
65	Action Sci-Fi Thriller


```

69 Action|Comedy|Horror|Thriller
70 Action
75 Action|Sci-Fi|Thriller
77 Action|Crime|Drama|Thriller
85 Action|Adventure|Drama

```

```

movies[is_action].head(15)

```

	movieId	title \
5	6	Heat (1995)
8	9	Sudden Death (1995)
9	10	GoldenEye (1995)
14	15	Cutthroat Island (1995)
19	20	Money Train (1995)
22	23	Assassins (1995)
41	42	Dead Presidents (1995)
43	44	Mortal Kombat (1995)
50	51	Guardian Angel (1994)
65	66	Lawnmower Man 2: Beyond Cyberspace (1996)
69	70	From Dusk Till Dawn (1996)
70	71	Fair Game (1995)
75	76	Screamers (1995)
77	78	Crossing Guard, The (1995)
85	86	White Squall (1996)

	genres
5	Action Crime Thriller
8	Action
9	Action Adventure Thriller
14	Action Adventure Romance
19	Action Comedy Crime Drama Thriller
22	Action Crime Thriller
41	Action Crime Drama
43	Action Adventure Fantasy
50	Action Drama Thriller
65	Action Sci-Fi Thriller
69	Action Comedy Horror Thriller
70	Action
75	Action Sci-Fi Thriller
77	Action Crime Drama Thriller
85	Action Adventure Drama

Group By and Aggregate

```

ratings_count = ratings[['movieId','rating']].groupby('rating').count()
ratings_count
movieId

```

rating	
0.5	239125
1.0	680732
1.5	279252
2.0	1430997
2.5	883398
3.0	4291193
3.5	2200156
4.0	5561926
4.5	1534824
5.0	2898660

```
average_rating = ratings[['movieId', 'rating']].groupby('movieId').mean()
average_rating.head()
```

	rating
movieId	
1	3.921240
2	3.211977
3	3.151040
4	2.861393
5	3.064592

```
movie_count = ratings[['movieId', 'rating']].groupby('movieId').count()
movie_count.head()
```

	rating
movieId	
1	49695
2	22243
3	12735
4	2756
5	12161

```
movie_count = ratings[['movieId', 'rating']].groupby('movieId').count()
movie_count.tail()
```

	rating
movieId	
131254	1
131256	1
131258	1
131260	1
131262	1

Merge Dataframes

```
tags.head()
```

userId	movieId	tag	timestamp
--------	---------	-----	-----------

0	18	4141	Mark Waters	2009-04-24 18:19:40
1	65	208	dark hero	2013-05-10 01:41:18
2	65	353	dark hero	2013-05-10 01:41:19
3	65	521	noir thriller	2013-05-10 01:39:43
4	65	592	dark hero	2013-05-10 01:41:18

```
movies.head()
```

	movieId	title \
0	1	Toy Story (1995)
1	2	Jumanji (1995)
2	3	Grumpier Old Men (1995)
3	4	Waiting to Exhale (1995)
4	5	Father of the Bride Part II (1995)

	genres
0	Adventure Animation Children Comedy Fantasy
1	Adventure Children Fantasy
2	Comedy Romance
3	Comedy Drama Romance
4	Comedy

```
t = movies.merge(tags, on='movieId', how='inner')
t.head()
```

	movieId	title	genres \
0	1	Toy Story (1995)	Adventure Animation Children Comedy Fantasy
1	1	Toy Story (1995)	Adventure Animation Children Comedy Fantasy
2	1	Toy Story (1995)	Adventure Animation Children Comedy Fantasy
3	1	Toy Story (1995)	Adventure Animation Children Comedy Fantasy
4	1	Toy Story (1995)	Adventure Animation Children Comedy Fantasy

	userId	tag	timestamp
0	1644	Watched	2014-12-04 23:44:40
1	1741	computer animation	2007-07-08 13:59:15
2	1741	Disney animated feature	2007-07-08 22:21:47
3	1741	Pixar animation	2007-07-08 22:46:10
4	1741	TÃ©a Leoni does not star in this movie	2009-06-15 19:19:33

Combine aggregation, merging, and filters to get useful analytics

Vectorized String Operations

```
movies.head()
```

	movieId	title \
0	1	Toy Story (1995)
1	2	Jumanji (1995)
2	3	Grumpier Old Men (1995)
3	4	Waiting to Exhale (1995)

4 5 Father of the Bride Part II (1995)

```

                                genres
0  Adventure|Animation|Children|Comedy|Fantasy
1                Adventure|Children|Fantasy
2                    Comedy|Romance
3                Comedy|Drama|Romance
4                    Comedy

```

Split 'genres' into multiple columns

```

movie_genres = movies['genres'].str.split('|', expand=True)
movie_genres[:10]

```

	0	1	2	3	4	5	6	7	8 \
0	Adventure	Animation	Children	Comedy	Fantasy	None	None	None	None
1	Adventure	Children	Fantasy	None	None	None	None	None	None
2	Comedy	Romance	None	None	None	None	None	None	None
3	Comedy	Drama	Romance	None	None	None	None	None	None
4	Comedy	None	None	None	None	None	None	None	None
5	Action	Crime	Thriller	None	None	None	None	None	None
6	Comedy	Romance	None	None	None	None	None	None	None
7	Adventure	Children	None	None	None	None	None	None	None
8	Action	None	None	None	None	None	None	None	None
9	Action	Adventure	Thriller	None	None	None	None	None	None

```

9
0  None
1  None
2  None
3  None
4  None
5  None
6  None
7  None
8  None
9  None

```

Add a new column for comedy genre flag

```

movie_genres['isComedy'] = movies['genres'].str.contains('Comedy')
movie_genres[:10]

```

	0	1	2	3	4	5	6	7	8 \
0	Adventure	Animation	Children	Comedy	Fantasy	None	None	None	None
1	Adventure	Children	Fantasy	None	None	None	None	None	None
2	Comedy	Romance	None	None	None	None	None	None	None
3	Comedy	Drama	Romance	None	None	None	None	None	None

4	Comedy	None	None	None	None	None	None	None	None
5	Action	Crime	Thriller	None	None	None	None	None	None
6	Comedy	Romance	None	None	None	None	None	None	None
7	Adventure	Children	None	None	None	None	None	None	None
8	Action	None	None	None	None	None	None	None	None
9	Action	Adventure	Thriller	None	None	None	None	None	None

```

9 isComedy
0 None      True
1 None      False
2 None      True
3 None      True
4 None      True
5 None      False
6 None      True
7 None      False
8 None      False
9 None      False

```

Extract year from title e.g. (2007)

```
movies['year'] = movies['title'].str.extract('.*\((.*)\)'.*, expand=True)
```

```
movies.tail()
```

	movieId	title	genres	year
27273	131254	Kein Bund für's Leben (2007)	Comedy	2007
27274	131256	Feuer, Eis & Dosenbier (2002)	Comedy	2002
27275	131258	The Pirates (2014)	Adventure	2014
27276	131260	Rentun Ruusu (2001)	(no genres listed)	2001
27277	131262	Innocence (2014)	Adventure Fantasy Horror	2014

Parsing Timestamps

```
tags.dtypes
```

```

userId      int64
movieId     int64
tag          object
timestamp    object
dtype: object

```

```
tags.head(5)
```

	userId	movieId	tag	timestamp
0	18	4141	Mark Waters	2009-04-24 18:19:40
1	65	208	dark hero	2013-05-10 01:41:18
2	65	353	dark hero	2013-05-10 01:41:19
3	65	521	noir thriller	2013-05-10 01:39:43
4	65	592	dark hero	2013-05-10 01:41:18

```
tags.head(2)
```

	userId	movieId	tag	timestamp
0	18	4141	Mark Waters	2009-04-24 18:19:40
1	65	208	dark hero	2013-05-10 01:41:18

Average Movie Ratings over Time

```
average_rating = ratings[['movieId', 'rating']].groupby('movieId', as_index=False).mean()
average_rating.tail()
```

	movieId	rating
26739	131254	4.0
26740	131256	4.0
26741	131258	2.5
26742	131260	3.0
26743	131262	4.0

```
joined = movies.merge(average_rating, on='movieId', how='inner')
joined.head
```

<bound method NDFrame.head of			movieId	title \
0	1		Toy Story (1995)	
1	2		Jumanji (1995)	
2	3		Grumpier Old Men (1995)	
3	4		Waiting to Exhale (1995)	
4	5		Father of the Bride Part II (1995)	
...	...		...	
26739	131254		Kein Bund für's Leben (2007)	
26740	131256		Feuer, Eis & Dosenbier (2002)	
26741	131258		The Pirates (2014)	
26742	131260		Rentun Ruusu (2001)	
26743	131262		Innocence (2014)	

	genres	year	rating
0	Adventure Animation Children Comedy Fantasy	1995	3.921240
1	Adventure Children Fantasy	1995	3.211977
2	Comedy Romance	1995	3.151040
3	Comedy Drama Romance	1995	2.861393
4	Comedy	1995	3.064592
...	...	...	...
26739	Comedy	2007	4.000000
26740	Comedy	2002	4.000000
26741	Adventure	2014	2.500000
26742	(no genres listed)	2001	3.000000
26743	Adventure Fantasy Horror	2014	4.000000

```
[26744 rows x 5 columns]>
```